

Tut 6

Ex. 9.7 Q11

$$\sum_{n=0}^{\infty} \frac{(-1)^n x^n}{n!}$$

Let $a_n = \frac{(-1)^n x^n}{n!}$ for $n \geq 0$.

$$\left| \frac{a_{n+1}}{a_n} \right| = \left| \frac{(-1)^{n+1} x^{n+1}}{(n+1)!} \cdot \frac{n!}{(-1)^n x^n} \right| = \left| (-1) x \cdot \frac{1}{n+1} \right| = \frac{|x|}{n+1}$$

$$\therefore \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \frac{|x|}{n+1} = |x| \lim_{n \rightarrow \infty} \frac{1}{n+1} = 0 \quad \forall x \in \mathbb{R}$$

Since $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = 0 < 1$ for all $x \in \mathbb{R}$, the series converges absolutely for all $x \in \mathbb{R}$. by ratio test.

(a) Radius of convergence = ∞ , interval of conv. = $(-\infty, \infty)$

(b) $(-\infty, \infty)$

(c) There are no values for which the series converges conditionally.

Q3)

$$\sum_{n=1}^{\infty} \frac{(4x-5)^{2n+1}}{n^{3/2}}$$

Let $a_n = \frac{(4x-5)^{2n+1}}{n^{3/2}}$ for $n \geq 1$.

$$\left| \frac{a_{n+1}}{a_n} \right| = \left| \frac{(4x-5)^{2n+3}}{(n+1)^{3/2}} \cdot \frac{n^{3/2}}{(4x-5)^{2n+1}} \right| = \left| (4x-5)^2 \cdot \left(\frac{n}{n+1} \right)^{3/2} \right|$$

$$= (4x-5)^2 \left(\frac{n}{n+1} \right)^{3/2}$$

$$\therefore \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} (4x-5)^2 \left(\frac{n}{n+1} \right)^{3/2} = (4x-5)^2 \lim_{n \rightarrow \infty} \left(\frac{n}{n+1} \right)^{3/2}$$

$$= (4x-5)^2$$

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| < 1 \Leftrightarrow (4x-5)^2 < 1 \Leftrightarrow |4x-5| < 1$$

$$\Leftrightarrow 1 < x < \frac{3}{2}$$

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| > 1 \Leftrightarrow |4x-5| > 1 \Leftrightarrow x < 1 \text{ or } x > \frac{3}{2}$$

For $1 < x < \frac{3}{2}$, the series conv. abs. by ratio test.

For $x < 1$ or $x > \frac{3}{2}$, the series div. by ratio test.

For $x = 1$, we have $\sum_{n=1}^{\infty} \frac{(-1)^{2n+1}}{n^{3/2}} = \sum_{n=1}^{\infty} \left(-\frac{1}{n^{3/2}}\right)$.

Since $\sum_{n=1}^{\infty} \left| -\frac{1}{n^{3/2}} \right| = \sum_{n=1}^{\infty} \frac{1}{n^{3/2}}$ converges by p-series ($\frac{3}{2} > 1$),

the series $\sum_{n=1}^{\infty} \left(-\frac{1}{n^{3/2}}\right)$ conv. abs.

For $x = \frac{3}{2}$, we have $\sum_{n=1}^{\infty} \frac{1}{n^{3/2}}$, which ~~is~~ converges abs.

(a) ...

(b) ...

(c) ...

Q34

$$\sum_{n=1}^{\infty} \frac{3 \cdot 5 \cdot 7 \cdots (2n+1)}{n^2 \cdot 2^n} x^{n+1}$$

$$\text{Let } a_n = \frac{3 \cdot 5 \cdot 7 \cdots (2n+1)}{n^2 \cdot 2^n} x^{n+1} \text{ for } n \geq 1.$$

$$\begin{aligned} \left| \frac{a_{n+1}}{a_n} \right| &= \left| \frac{3 \cdot 5 \cdot 7 \cdots (2n+3)}{(n+1)^2 \cdot 2^{n+1}} \cdot \frac{n^2 \cdot 2^n}{3 \cdot 5 \cdot 7 \cdots (2n+1)} \cdot \frac{x^{n+2}}{x^{n+1}} \right| \\ &= \frac{(2n+3)n^2}{(n+1)^2 \cdot 2} |x| \end{aligned}$$

$$\text{Since } \lim_{n \rightarrow \infty} \frac{(2n+3)n^2}{2(n+1)^2} = \infty \quad (\text{can use L'Hôpital}),$$

we have $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \infty$ for all $x \neq 0$. $\therefore$ The series diverges by ratio test.

For $x=0$, $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = 0$. $\therefore$ The series converges absolutely by ratio test.

(a) Radius of conv. = 0, interval of conv. = 0

(b) $x=0$

(c) ...

Q40

$$\sum_{n=1}^{\infty} \left(\frac{n}{n+1}\right)^{n^2} x^n$$

Let $a_n = \left(\frac{n}{n+1}\right)^{n^2} x^n$ for $n \geq 1$.

$$\sqrt[n]{|a_n|} = \sqrt[n]{\left|\left(\frac{n}{n+1}\right)^{n^2} x^n\right|} = \sqrt[n]{\left(\frac{n}{n+1}\right)^{n^2} |x|^n} = \left(\frac{n}{n+1}\right)^n |x|$$

$$\begin{aligned}\therefore \lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} &= \lim_{n \rightarrow \infty} \left(\frac{n}{n+1}\right)^n |x| = |x| \lim_{n \rightarrow \infty} \left(\frac{1}{1+\frac{1}{n}}\right)^n \\ &= |x| \lim_{n \rightarrow \infty} \frac{1}{\left(1+\frac{1}{n}\right)^n} \\ &= \frac{|x|}{e}\end{aligned}$$

$$\begin{aligned}\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} < 1 &\Leftrightarrow \frac{|x|}{e} < 1 \Leftrightarrow |x| < e \quad \therefore \text{For } |x| < e, \text{ the series} \\ &\quad \text{converges absolutely, by root test.} \\ \lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} > 1 &\Leftrightarrow \frac{|x|}{e} > 1 \Leftrightarrow |x| > e \quad \therefore \text{For } |x| > e, \text{ the series} \\ &\quad \text{diverges, by root test.}\end{aligned}$$

$\therefore \text{Radius of conv.} = e$

Q42

$$\sum_{n=0}^{\infty} (e^x - 4)^n$$

Let $a_n = (e^x - 4)^n$ for $n \geq 1$.

$$\sqrt[n]{|a_n|} = |e^x - 4|$$

$$\therefore \lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} = |e^x - 4|$$

$$\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} < 1 \Leftrightarrow |e^x - 4| < 1 \Leftrightarrow -1 < e^x - 4 < 1$$

$$\Leftrightarrow 3 < e^x < 5$$

$$\Leftrightarrow \ln 3 < \ln e^x < \ln 5$$

$$\Leftrightarrow \ln 3 < x < \ln 5 \quad \because \ln e^x = x$$

$\therefore$ For $\ln 3 < x < \ln 5$, the series conv. abs. by root test.

For $x = \ln 3$, we have $\sum_{n=0}^{\infty} (e^{\ln 3} - 4)^n = \sum_{n=0}^{\infty} (-1)^n$ which div.

For $x = \ln 5$, we have $\sum_{n=0}^{\infty} (e^{\ln 5} - 4)^n = \sum_{n=0}^{\infty} 1$ which div. $\left(\lim_{n \rightarrow \infty} 1 \neq 0 \right.$
and by n-term test)

∴ The interval of conv. is $\ln 3 < x < \ln 5$.

For $\ln 3 < x < \ln 5$,

$\sum_{n=0}^{\infty} (e^x - 4)^n$ is a geometric series.

$$\therefore \sum_{n=0}^{\infty} (e^x - 4)^n = \frac{1}{1 - (e^x - 4)} = \frac{1}{5 - e^x}.$$